Question 1:

(a) Convert the following decimal value to binary and truncate at 4 bits after the radix. (9 Points)

$$44.2025_{10}$$

(c) Perform the following multiplication in binary (base-2) and convert the result back into decimal. Recall: convert each operand to base-2 and reformulate the expression before performing the operation. (10 Points)

$$22_{10} \times 11_{10}$$

Question 2:

(a) Simplify each of the following expressions in one step by identifying and using a single theorem/law of Boolean algebra. Recall: expressions inside parentheses can be treated as whole variables listed in your laws/theorems. (15 Points)

i) $E'(A + F')DEF$

ii) $(A'D + C)B'C$

iii) $(ABC' + D)(ABC' + EF')$

(b) Prove the following equation. (15 Points)

Hint: suggested approach - use the consensus theorem to add a term before simpifying RHS.

$$A'BC + A'C'D + AD = C'D + A'BC + AD + BD$$

Question 3:

Use the following function for all parts of this question:

$$F(T,X,Y,Z) = (XZ' \oplus Y)'(XYZ' + TY)$$

(a) Draw a circuit for the expression as it is given, without changing/simplifying. (8 Points)

(b) Eliminate the XOR and simplify to obtain the minimum SOP of the function, then draw a circuit implementation for the simplified function. Hint: simplify to 2 terms. (12 Points)

$$(XZ' \oplus Y)'(XYZ' + TY)$$

(c) Is the following statement always true? Explain your reasoning. (5 Points)

" If $AB + C' = C' + D$, then $AB = D$ "

Question 4:

Consider the following situation for all parts of this question:
In an organization meeting, a proposal can be adopted if it: receives votes from a majority of its members or all of its officers, does not violate their rules, and does not exceed the budget.

(a) Identify all input variables and convert the statement into a Boolean function. (8 Points)

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members or all of its officers, does not violate their rules, and does not exceed the budget.

(b) Find the minterm expansion of the function; give your final answer in SOP form. (8 Points)

(c) Find the maxterm expansion of the function. (9 Points)